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MODUL 12

Laravel : Database 1

1. Konfigurasi Environment (.env)

- a. Code (boleh copy/screenshot dengan syarat terlihat jelas dan tidak blur)

Modifikasi pada bagian koneksi ke MySQL/MariaDB:

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=ecommerce
DB_USERNAME=root
DB_PASSWORD=
```

2. File Migration (database/migrations/..._create_products_table.php)

Mendefinisikan skema secara aman.

- a. Code (boleh copy/screenshot dengan syarat terlihat jelas dan tidak blur)

```
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     *
     * @return void
     */
    public function up()
```

```

{
    Schema::create('products', function (Blueprint
$table) {
        $table->id();
        $table->string('name');
        $table->integer('price');
        $table->timestamps();
    });
}

/**
 * Reverse the migrations.
 *
 * @return void
 */
public function down()
{
    Schema::dropIfExists('products');
}
};

```

3. Model Eloquent (app/Models/Product.php)

- a. Code (boleh copy/screenshot dengan syarat terlihat jelas dan tidak blur)

Penentuan perlindungan Mass Assignment.

```

<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Product extends Model
{
    use HasFactory;

    protected $fillable = ['name', 'price'];
}

```

4. Controller (app/Http/Controllers/ProductController.php)

a. Code (boleh copy/screenshot dengan syarat terlihat jelas dan tidak blur)

```
<?php

namespace App\Http\Controllers;

use App\Models\Product;
use Illuminate\Http\Request;

class ProductController extends Controller
{
    public function index()
    {
        $products = Product::all();
        return view('products.index', ['products' => $products]);
    }

    public function create()
    {
        return view('products.form', [
            'title' => 'Tambah',
            'product' => new Product(),
            'route' => route('products.store'),
            'method' => 'POST',
        ]);
    }

    public function store(Request $request)
    {
        $validated = $request->validate([
            'name' => 'required|min:4',
            'price' => 'required|integer|min:1000000',
        ]);

        Product::create($validated);
        return
        redirect()->route('products.index')->with('success', 'Produk
```

```

        berhasil ditambahkan');
    }

    public function show(Product $product)
    {
        return view('products.show', compact('product'));
    }

    public function edit(Product $product)
    {
        return view('products.form', [
            'title' => 'Edit',
            'product' => $product,
            'route' => route('products.update', $product),
            'method' => 'PUT',
        ]);
    }

    public function update(Request $request, Product $product)
    {
        $validated = $request->validate([
            'name' => 'required|min:4',
            'price' => 'required|integer|min:1000000',
        ]);

        $product->update($validated);
        return
        redirect()->route('products.index')->with('success', 'Produk
        berhasil diperbarui');
    }

    public function destroy(Product $product)
    {
        $product->delete();
        return
        redirect()->route('products.index')->with('success', 'Produk
        berhasil dihapus');
    }
}

```

5. Daftarkan Routing (routes/web.php)

- a. Code (boleh copy/screenshot dengan syarat terlihat jelas dan tidak blur)

```
<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\ProductController;

/*
| -----
|
| Web Routes
| -----
|
| Here is where you can register web routes for your
| application. These
| routes are loaded by the RouteServiceProvider within a
| group which
| contains the "web" middleware group. Now create something
| great!
|
*/

Route::get('/', function () {
    return view('welcome');
});

Route::resource('products', ProductController::class);
```

6. Pembuatan View (Tampilan CRUD)

- a. Membuat File View Index (resources/views/products/index.blade.php)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width,
initial-scale=1.0">
<title>Daftar Produk</title>
<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/b
ootstrap.min.css" rel="stylesheet">
</head>
<body class="container">
<div class="row justify-content-center mt-5">
<div class="col-md-8">
@if(session('success'))
<div class="alert alert-success">{{
session('success') }}</div>
@endif
<div class="d-flex justify-content-between
align-items-center mb-3">
<span>{{ session('msg') }}</span>
<a href="{{ route('products.create') }}"
class="btn btn-primary">Tambah</a>
</div>
<table class="table table-bordered
table-striped">
<thead>
<tr>
<th>Nama</th>
<th>Harga</th>
<th>Aksi</th>
</tr>
</thead>
<tbody>
@foreach($products as $product)
<tr>
<td>{{ $product->name }}</td>
<td>{{ $product->price }}</td>
<td>
<a href="{{
route('products.edit', $product->id) }}" class="btn btn-sm
btn-primary">Edit</a>
<form method="POST" action="{{
```

```

route('products.destroy', $product->id) }}"
style="display:inline" onsubmit="return confirm('Yakin
hapus?') ">

                                @csrf
                                @method('DELETE')
                                <button class="btn btn-sm
btn-danger">Hapus</button>

                                </form>
                                </td>
                                </tr>
                                @endforeach
                                </tbody>
                                </table>
                                </div>
                                </div>
                                <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/boo
tstrap.bundle.min.js"></script>
</body>
</html>

```

b. Membuat File View Form (resources/views/products/form.blade.php)

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,
initial-scale=1.0">
    <title>Form {{ $title }} Produk</title>
    <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/b
ootstrap.min.css" rel="stylesheet">
</head>
<body class="container">
    <div class="row justify-content-center mt-5">
        <div class="col-md-6">
            <h4>Form {{ $title }} Produk</h4>
            <form class="border p-4" method="POST" action="{{

```

```

$route }}">

        @csrf
        @if($method === 'PUT')
            @method('PUT')
        @endif
        <div class="mb-3">
            <label for="name"
class="form-label">Nama</label>
            <input type="text" name="name" id="name"
class="form-control @error('name') is-invalid @enderror"
value="{{ old('name', $product->name) }}">
            @error('name')
                <div class="invalid-feedback">{{
$message }}</div>
            @enderror
        </div>
        <div class="mb-3">
            <label for="price"
class="form-label">Harga</label>
            <input type="number" name="price"
id="price" class="form-control @error('price') is-invalid
@enderror" value="{{ old('price', $product->price) }}">
            @error('price')
                <div class="invalid-feedback">{{
$message }}</div>
            @enderror
        </div>
        <div class="text-center">
            <button type="submit" class="btn
btn-success">Simpan</button>
        </div>
    </form>
</div>
</div>
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/boo
tstrap.bundle.min.js"></script>
</body>
</html>

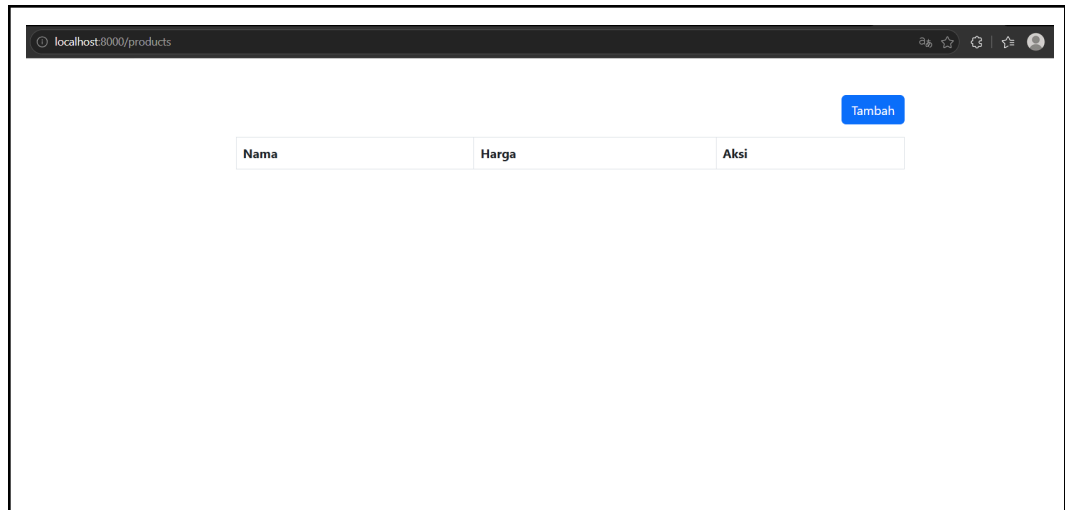
```


c. Membuat File View Show (resources/views/products/show.blade.php)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Detail Produk</title>
  <link
href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.0/dist/css/b
ootstrap.min.css" rel="stylesheet">
</head>
<body class="container">
  <div class="row justify-content-center mt-5">
    <div class="col-md-6">
      <h4>Detail Produk</h4>
      <div class="card p-4">
        <p><strong>Nama:</strong> {{ $product->name
}}</p>
        <p><strong>Harga:</strong> {{ $product->price
}}</p>
        <a href="{{ route('products.index') }}"
class="btn btn-secondary">Kembali</a>
      </div>
    </div>
  </div>
</body>
</html>
```

d. Screenshot output

1. Halaman Daftar Produk Awal (Kosong):



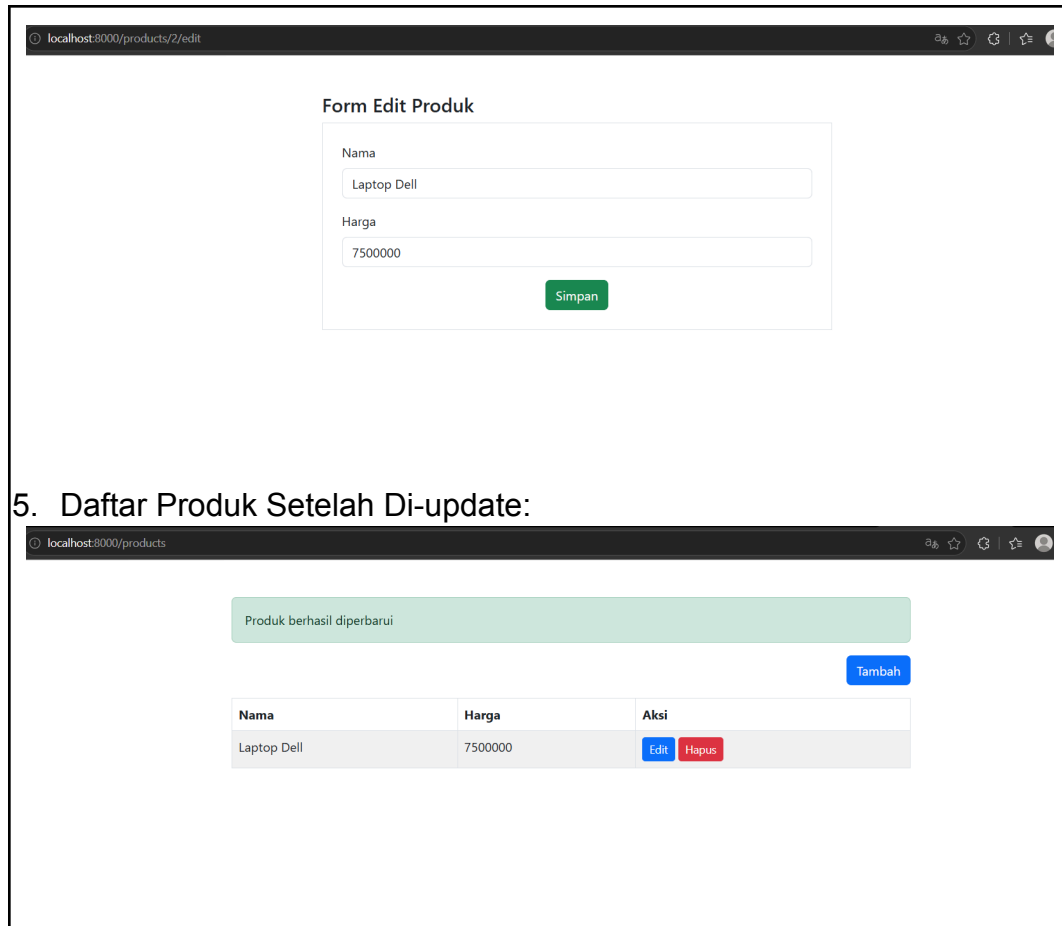
2. Form Tambah Produk (Mengisi Data):

The screenshot shows a web browser window with the address bar displaying 'localhost:8000/products/create'. The page content includes a form titled 'Form Tambah Produk'. The form has two input fields: 'Nama' with the value 'Laptop Dell' and 'Harga' with the value '9.000000'. A green button labeled 'Simpan' is located below the input fields.

3. Daftar Produk Setelah Ditambahkan (Sukses):

The screenshot shows a web browser window with the address bar displaying 'localhost:8000/products'. The page content includes a green message box that says 'Produk berhasil ditambahkan'. Below the message box is a table with three columns: 'Nama', 'Harga', and 'Aksi'. The table contains one row with the values 'Laptop Dell' and '9000000'. The 'Aksi' column contains two buttons: 'Edit' (blue) and 'Hapus' (red). A blue button labeled 'Tambah' is positioned in the top right corner of the table area.

4. Halaman Edit/Ubah Data Produk:



7. Templating Halaman (Blade @extends dan @yield)

a. Membuat File Template Utama (resources/views/template.blade.php)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>@yield('title')</title>
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/b
ootstrap.min.css" rel="stylesheet">
  <script src="{{ asset('js/jquery.min.js') }}"></script>
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/boo
```

```

tstrap.bundle.min.js"></script>
</head>
<body style="width:95%" class="container">
    @yield('content')
</body>
</html>

```

b. Mengubah View Index (resources/views/products/index.blade.php)

```

@extends('template')

@section('title', 'Daftar Produk')

@section('content')
<body class="container">
    <div class="row justify-content-center mt-5">
        <div class="col-md-8">
            @if(session('success'))
                <div class="alert alert-success">{{
session('success') }}</div>
            @endif
            <div class="d-flex justify-content-between
align-items-center mb-3">
                <span>{{ session('msg') }}</span>
                <a href="{{ route('products.create') }}"
class="btn btn-primary">Tambah</a>
            </div>
            <table class="table table-bordered
table-striped">
                <thead>
                    <tr>
                        <th>Nama</th>
                        <th>Harga</th>
                        <th>Aksi</th>
                    </tr>
                </thead>
                <tbody>
                    @foreach($products as $product)
                        <tr>

```

```

                <td>{{ $product->name }}</td>
                <td>{{ $product->price }}</td>
                <td>
                    <a href="{{
route('products.edit', $product->id) }}" class="btn btn-sm
btn-primary">Edit</a>
                    <form method="POST" action="{{
route('products.destroy', $product->id) }}"
style="display:inline" onsubmit="return confirm('Yakin
hapus?') ">
                        @csrf
                        @method('DELETE')
                        <button class="btn btn-sm
btn-danger">Hapus</button>
                    </form>
                </td>
            </tr>
        @endforeach
    </tbody>
</table>
</div>
</div>

<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/boo
tstrap.bundle.min.js"></script>
</body>
@endsection

```

c. Mengubah View Form (resources/views/products/form.blade.php)

```

@extends('template')

@section('title', 'Form ' . $title . ' Produk')

@section('content')
<body class="container">
    <div class="row justify-content-center mt-5">
        <div class="col-md-6">

```

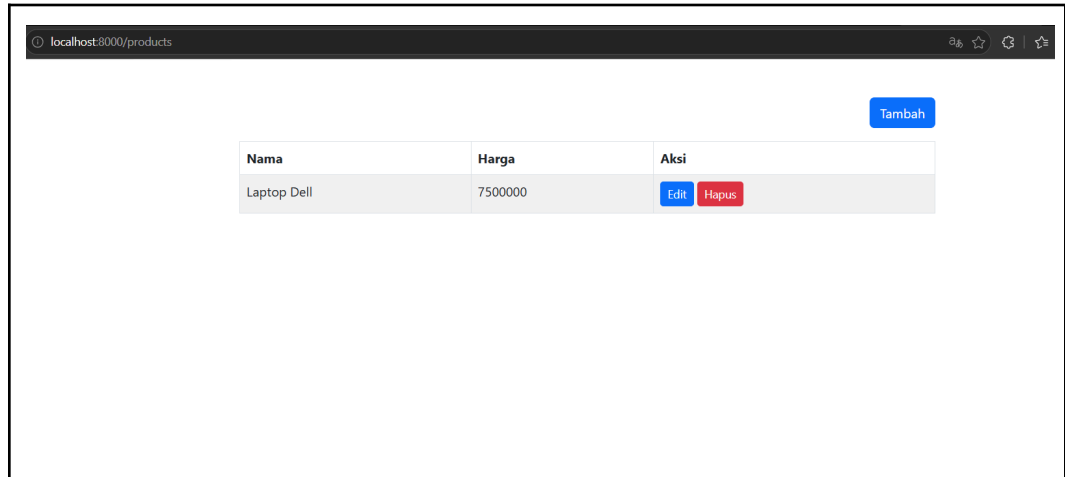
```

        <h4>Form {{ $title }} Produk</h4>
        <form class="border p-4" method="POST" action="{{
$route }}">
            @csrf
            @if($method === 'PUT')
                @method('PUT')
            @endif
            <div class="mb-3">
                <label for="name"
class="form-label">Nama</label>
                <input type="text" name="name" id="name"
class="form-control @error('name') is-invalid @enderror"
value="{{ old('name', $product->name) }}">
                @error('name')
                    <div class="invalid-feedback">{{
$message }}</div>
                @enderror
            </div>
            <div class="mb-3">
                <label for="price"
class="form-label">Harga</label>
                <input type="number" name="price"
id="price" class="form-control @error('price') is-invalid
@enderror" value="{{ old('price', $product->price) }}">
                @error('price')
                    <div class="invalid-feedback">{{
$message }}</div>
                @enderror
            </div>
            <div class="text-center">
                <button type="submit" class="btn
btn-success">Simpan</button>
            </div>
        </form>
    </div>
</div>
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/boo
tstrap.bundle.min.js"></script>

```

```
</body>  
@endsection
```

d. Screenshot output



8. Form Validation (Validasi Form Sisi Server)

a. Penjelasan Logika Validasi di Controller (ProductController.php)

Aturan validasi didefinisikan pada method store() dan update() sebelum data disimpan ke database:

```
$validated = $request->validate([  
    'name' => 'required|min:4',  
    'price' => 'required|integer|min:1000000',  
]);
```

Penjelasan Aturan:

1. 'name' => 'required|min:4': Field Nama wajib diisi dan panjang karakter minimal 4 karakter.
2. 'price' => 'required|integer|min:1000000': Field Harga wajib diisi, bertipe integer (angka), dan bernilai minimal 1.000.000 (satu juta rupiah).

b. Penjelasan Directive Validasi di View (form.blade.php)

Di file form.blade.php, kita menggunakan directive `@error` bawaan Blade untuk mendeteksi apakah terjadi error validasi pada field tertentu, kemudian menampilkan pesan error-nya:

```
<div class="mb-3">
    <label for="name"
class="form-label">Nama</label>
    <input type="text" name="name" id="name"
class="form-control @error('name') is-invalid @enderror"
value="{{ old('name', $product->name) }}">
    @error('name')
        <div class="invalid-feedback">{{
$message }}</div>
    @enderror
</div>
<div class="mb-3">
    <label for="price"
class="form-label">Harga</label>
    <input type="number" name="price"
id="price" class="form-control @error('price') is-invalid
@enderror" value="{{ old('price', $product->price) }}">
    @error('price')
        <div class="invalid-feedback">{{
$message }}</div>
    @enderror
</div>
```

`old('name')` digunakan agar ketika form gagal divalidasi, data lama yang diinputkan pengguna tidak hilang (ter-repopulate).

c. Hasil Pengujian Validasi (Screenshot):

Ketika kita mencoba menginputkan data yang tidak valid, misalnya:

1. Nama: Abc (hanya 3 karakter, kurang dari batas minimal yaitu 4)
2. Harga: 5000 (kurang dari batas minimal yaitu 1.000.000)

Sistem secara otomatis menolak penyimpanan data, memunculkan outline merah (is-invalid), dan menampilkan pesan kesalahan berikut:

127.0.0.1:8000/products/create

Form Tambah Produk

Nama

abc

The name must be at least 4 characters.

Harga

5000

The price must be at least 100000.

Simpan